

UNIT-WISE IN-CLASS ASSIGNMENT PACK

FUNDAMENTAL C++ FOR ROBOTICS (ETCCCPL205)

COURSE DETAILS

- **School:** School of Engineering & Technology
 - **Program:** B.Tech (Robotics)
 - **Course Name:** Fundamental C++ for Robotics
 - **Course Code:** ETCCCPL205
 - **Assignment Type:** In-Class Assignments (Unit-wise)
 - **Duration (Each):** 50 Minutes
-

PURPOSE OF IN-CLASS ASSIGNMENTS

The in-class assignments are designed to: - Reinforce **conceptual clarity** immediately after teaching - Encourage **logical reasoning and coding under time constraints** - Assess **Bloom's levels L2-L4** progressively - Prepare students for labs, assignments, and end-term evaluations

Each unit contains **one structured in-class assignment**, following the **approved institutional format**.

UNIT-1 IN-CLASS ASSIGNMENT

C++ Foundations & Control Logic

Bloom's Level: L2-L3 (Understand-Apply)

CO Mapping: CO1

Task A: Quick Concept Check (Answer in one line)

1. What is the purpose of the `main()` function?
 2. Difference between compiler and interpreter.
 3. One robotics example where `if-else` is required.
 4. Why are loops important in robot motion control?
-

Task B: Logic & Dry Run

A robot takes distance input from a sensor.

Distance (cm)	Action
< 20	STOP
20–50	SLOW DOWN
> 50	MOVE FORWARD

1. Which control structure is suitable and why?
2. Write the output for input distances: 15, 35, 70.

Task C: Coding Task

Write a C++ program to implement the above robot decision logic using `if-else`.

Exit Ticket

Write one thing you understood clearly today.

UNIT-2 IN-CLASS ASSIGNMENT

Memory Management & STL for Robotics

Bloom's Level: L3-L4 (Apply-Analyze)

CO Mapping: CO2

Task A: Concept Recall

1. Difference between stack and heap.
2. What is a pointer?
3. Why is STL preferred in robotics software?

Task B: Analytical Reasoning

A robot stores last 10 sensor readings.

1. Which STL container is best and why?
2. What happens if dynamic memory is not deallocated?

Task C: Coding Task

Write a C++ program that: - Uses a `vector` to store sensor values - Prints the average sensor reading

Exit Ticket

Mention one memory issue that can crash a robot program.

UNIT-3 IN-CLASS ASSIGNMENT

Object-Oriented Design for Robotics

Bloom's Level: L3-L4 (Apply-Analyze)

CO Mapping: CO3

Task A: Concept Check

1. What is encapsulation?
 2. Difference between inheritance and composition.
 3. Why polymorphism is useful in robotics?
-

Task B: Design Thinking

Design a robot system with **Sensor**, **Controller**, and **Motor**.

1. Identify which relationship is *has-a*.
 2. Identify which relationship is *is-a*.
-

Task C: Coding Task

Write a C++ class `Motor` with: - speed as private data - functions to start and stop the motor

Exit Ticket

Write one advantage of OOP in robotics software.

UNIT-4 IN-CLASS ASSIGNMENT

Embedded & Real-Time Programming

Bloom's Level: L4-L5 (Analyze-Evaluate)

CO Mapping: CO4, CO5

Task A: Conceptual Questions

1. What is latency in robotics?
 2. Why is deterministic timing important?
 3. What is a control loop?
-

Task B: Evaluation Scenario

A control loop is designed to run every 50 ms but sometimes takes 80 ms.

1. Is this system safe? Why or why not?
 2. What could be the reason for delay?
-

Task C: Pseudo-Code Task

Write pseudo-code for a control loop that: - Reads sensor - Makes decision - Waits for next cycle

Exit Ticket

Mention one risk of missing a control loop deadline.

GENERAL EVALUATION RUBRIC (Each Unit – 10 Marks)

Criteria	Marks
Concept clarity	3
Logical reasoning	3
Coding / Pseudo-code	3
Exit ticket	1
Total	10

End of Unit-wise In-Class Assignment Pack